EPP'S THEOREM: A FINITE WEAK SOLUTION EXISTS FOR AN EXTENSION OF DISCRETE VALUATION RINGS WHEN THE P-POWER CORE OF THE BIGGER RESIDUE FIELD IS SEPARABLE OVER THE SMALLER

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Allowed standard prerequisites (editorial).

- Abhyankar's lemma on ramification and the construction of extensions adjoining a root of a uniformizer. invoked at proof lines 35533-35543. Settles the tame case and supplies the formal smoothness conclusion used there.
- Strict henselization of a discrete valuation ring and its effect on the residue field. invoked at proof lines 35547-35555. Is the first reduction, replacing the residue fields by their separable closures.
- Completions of discrete valuation rings, formal smoothness and the descent of solutions. used at proof lines 35558-35586. Is the second reduction, moving the problem to the complete case and then descending the answer.
- The Cohen structure theorem and Cohen rings. invoked at proof lines 35600-35605. Produces the ring Lambda mapping to A with t sent to pi, which the explicit construction in the complete case uses.
- Weak solutions for an extension of discrete valuation rings. the definition invoked at statement lines 35526-35527. Fixes the property that K_1/K is required to have.

Source reference. Original lemma 031D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3980–3980. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 032A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4013–4013. Cross-chapter navigation pointer into

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the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 34445: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34445–34445. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 34037: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34037–34037. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 33814: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33814–33814. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 32989: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32989–32989. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 34006: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34006–34006. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 9257: sources/stacks-more-algebra/pinned-original-source.tex, source lines 9257–9257. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 33954: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33954–33954. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 34555: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34555–34555. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 34517: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34517–34517. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 35370: sources/stacks-more-algebra/pinned-original-source.tex, source lines 35370–35370. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 33882: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33882–33882. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Theorem 0.2 (Epp). *Let $A \subset B$ be an extension of discrete valuation rings with fraction fields $K \subset L$. If the characteristic of κ_A is $p > 0$, assume that every element of*

$$\bigcap\nolimits_{n\geq 1} \kappa_B^{p^n}$$

is separable algebraic over κ_A. Then there exists a finite extension K_1/K which is a weak solution for $A \to B$ as defined in Definition 34445.

Proof. If the characteristic of κ_A is zero or if the residue characteristic is p, the ramification index is prime to p, and the residue field extension is separable, then this follows from Abhyankar's lemma (Lemma 34037). Namely, suppose the ramification index is e. Choose a uniformizer $\pi \in A$. Let K_1/K be the extension obtained by adjoining an eth root of π. By Lemma 33954 we see that the integral closure A_1 of A in K_1 is a discrete valuation ring with ramification index over A. Thus $A_1 \to (B_1)_{\mathfrak{m}}$ is formally smooth in the $\mathfrak{m}$-adic topology for all maximal ideals $\mathfrak{m}$ of B_1 by Lemma 34037 and a fortiori these are weakly unramified extensions of discrete valuation rings.

From now on we let p be a prime number and we assume that κ_A has characteristic p. We first apply Lemma 34555 to reduce to the case that A and B have separably closed residue fields. Since κ_A and κ_B are replaced by their separable algebraic closures by this procedure we see that we obtain

$$\kappa_A \supset \bigcap\nolimits_{n\geq 1} \kappa_B^{p^n}$$

from the condition of the theorem.

Let $\pi \in A$ be a uniformizer. Let $A^\wedge$ and $B^\wedge$ be the completions of A and B. We have a commutative diagram

$$\begin{array}{ccc} B & \longrightarrow & B^\wedge \\ \uparrow & & \uparrow \\ A & \longrightarrow & A^\wedge \end{array}$$

of extensions of discrete valuation rings. Let $K^\wedge$ be the fraction field of $A^\wedge$. Suppose that we can find a finite extension $M/K^\wedge$ which is (a) a weak solution for $A^\wedge \to B^\wedge$ and (b) a compositum of a separable extension and an extension obtained by adjoining a p-power root of π. Then by Lemma 33814 we can find a finite extension K_1/K such that $K^\wedge \otimes_K K_1 = M$. Let A_1, resp. $A_1^\wedge$ be the integral closure of A, resp. $A^\wedge$ in K_1, resp. M. Since $A \to A^\wedge$ is formally smooth in the $\mathfrak{m}^\wedge$-adic topology (Lemma 32989) we see that $A_1 \to A_1^\wedge$ is formally smooth in the $\mathfrak{m}_1^\wedge$-adic topology (Lemma 34006 and A_1 and $A_1^\wedge$ are discrete valuation rings by discussion in Remark 33882). We conclude from Lemma 34517 part (2) that K_1/K is a weak solution for $A \to B^\wedge$. Applying Lemma 34517 part (1) we see that K_1/K is a weak solution for $A \to B$.

Thus we may assume A and B are complete discrete valuation rings with separably closed residue fields of characteristic p and with $\kappa_A \supset \bigcap_{n\geq 1} \kappa_B^{p^n}$. We are also given a uniformizer $\pi \in A$ and we have to find a weak solution for $A \to B$ which is a compositum of a separable extension and a field obtained by taking p-power roots of π. Note that the second condition is automatic if A has mixed characteristic.

Set $k = \bigcap_{n\geq 1} \kappa_B^{p^n}$. Observe that k is an algebraically closed field of characteristic p. If A has mixed characteristic let Λ be a Cohen ring for k and in the equicharacteristic case set $\Lambda = k[[t]]$. We can choose a ring map $\Lambda \to A$ which maps t to π in the equicharacteristic case. In the equicharacteristic case this follows from the Cohen structure theorem (Algebra, Theorem 032A) and in the mixed characteristic case

this follows as $\mathbf{Z}_p \to \Lambda$ is formally smooth in the adic topology (Lemmas 32989 and 9257). Applying Lemma 34517 we see that it suffices to prove the existence of a weak solution for $\Lambda \to B$ which in the equicharacteristic p case is a compositum of a separable extension and a field obtained by taking p-power roots of t. However, since $\Lambda = k[[t]]$ in the equicharacteristic case and any extension of $k((t))$ is such a compositum, we can now drop this requirement!

Thus we arrive at the situation where A and B are complete, the residue field k of A is algebraically closed of characteristic $p > 0$, we have $k = \bigcap \kappa_B^{p^n}$, and in the mixed characteristic case p is a uniformizer of A (i.e., A is a Cohen ring for k). If A has mixed characteristic choose a Cohen ring Λ for κ_B and in the equicharacteristic case set $\Lambda = \kappa_B[[t]]$. Arguing as above we may choose a ring map $A \to \Lambda$ lifting $k \to \kappa_B$ and mapping a uniformizer to a uniformizer. Since $k \subset \kappa_B$ is separable the ring map $A \to \Lambda$ is formally smooth in the adic topology (Lemma 32989). Hence we can find a ring map $\Lambda \to B$ such that the composition $A \to \Lambda \to B$ is the given ring map $A \to B$ (see Lemma 9257). Since Λ and B are complete discrete valuation rings with the same residue field, B is finite over Λ (Algebra, Lemma 031D). This reduces us to the special case discussed in Lemma 35370. $\square$